

K-FIELD

SPECTROPHOTOMETER-KEPLER COMPARISON

A Unified Spatial-Phase, Reciprocal-Geometry, and Traversal-Kinematics Framework

Textbook-quality archival reference for the related terrestrial spectrophotometer and Kepler spacecraft experiments

Field	Value
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This reference presents measured observations, derived geometry, internal mathematical closure, cross-experiment recurrence, inferred structure, unresolved dimensions, and explicit reproduction equations. All numerical values and figures are taken from the identified source files or calculated directly from those values.

Abstract

This archival reference develops a unified comparison of two direction-sensitive experimental systems: a rotating multichannel spectrophotometer carried by the rotating and translating Earth, and a two-axis pixel-variance analysis of the Kepler spacecraft trajectory. The terrestrial pic_id14C record contains 84,330 measurements, 937 complete 90-position rotations, and 933 five-turn profiles over 11.97 hours. A blue-versus-lower-energy/NIR spectral mode exhibits a changing crossing period that is represented by one inertial wavevector. The resolved normal is RA 291.732 deg, Dec +35.972 deg, with primitive spacing 1,566.144 km. Projected velocity explains the observed chirp, and phase-derived distance agrees with integrated normal velocity to 1.682 m after 38,592.740 km of projected travel. The extended 24.064-hour pic_id14E record exposes two independently recovered CMB-transverse reciprocal directions, qA and qB. Their cross product is 0.036386 deg from the supplied CMB axis, and their direct dual defines a base face with 319.393 km and 1,039.778 km edges, 95.609 deg included angle, and 330,508.374 km² area. The earlier q14C wavevector lies only 0.084802 deg from the qA-qB plane and is reconstructed as $-0.0332522 \text{ qA} + 0.6624680 \text{ qB}$ with 0.1480 percent relative residual. Independently, the Kepler translated direct-lattice solution contains a low-order reciprocal family indexed by (1,-2,-3) whose refined normal is 0.640396 deg from q14C in heliocentric J2000 ecliptic coordinates. Kepler also exposes separate chirality and sheet-array branches, including a 228,650 km modular period, a 655,343 km sheet interval, and 2,292 alternating chirality reversals. The combined result is a multi-axis spatial-phase architecture in which detector-specific states sample fixed reciprocal directions at temporal rates set by observer velocity.

Source basis: S1-S9

Executive Overview

What was investigated. The comparison asks whether two very different instruments can be described by the same minimum spatial organization. The spectrophotometer records wavelength-resolved intensity while a device rotates in a laboratory fixed to Earth. Kepler records pixel-level optical variance while a spacecraft moves along a heliocentric orbit. The instruments, epochs, data products, trajectories, and measured states are different. The comparison therefore focuses on invariant geometry and kinematics rather than waveform identity.

What was found. The terrestrial record resolves a reciprocal frame transverse to the supplied CMB translation direction. A shorter interval first exposed one wavevector, q14C. The longer interval later exposed two basis directions, qA and qB, and showed that q14C is a mixed coordinate within their plane. The Kepler direct-lattice analysis independently returns a low-order reciprocal plane whose normal falls in the same directional sector. Kepler also returns other axes and larger spatial periods, showing that the common direction is one branch of a larger multi-axis system rather than the only axis present.

Why the trajectory matters. A fixed spatial spacing does not generally appear as a fixed temporal period. The phase rate is the dot product of wavevector and observer velocity. The spectrophotometer directly demonstrates this through a contracting modulation period and a changing Earth-rotation contribution. The Kepler chirality ledger demonstrates the complementary form: the time between eligible state updates is quantized in spacecraft cadence units, while the spatial distance per cadence changes smoothly because projected velocity changes.

What the 1.68 m closure means. The raw spectrophotometer signal supplies the measured chirp, spectral phase groups, channel responses, and reversal sequence. Those observations determine a spatial coordinate. Within that coordinate, phase distance and integrated velocity close to 1.682 m after 38.59 million metres. This value is not a direct pixel or spectrometer ranging resolution; it is the numerical closure of the signal-derived geometry and trajectory representation.

What is now established within the supplied corpus. The combined data require a representation with fixed reciprocal directions, position-dependent phase, observer-dependent phase rate, detector-specific transduction, multiple axes, and hierarchical spatial scales. A single closed laboratory circle, a purely time-periodic oscillator, or one universal plane spacing is insufficient to contain the full set of observations.

What remains open. The terrestrial 24-hour trajectory resolves a two-dimensional base face but not the longitudinal direct-space period. The terrestrial and Kepler absolute phase origins have not been registered in one Sun-centered ledger. The microscopic mechanism converting spatial phase into wavelength redistribution or pixel-variance chirality is also not yet specified.

Evidence and derivation graph linking the spectrophotometer and Kepler experiments

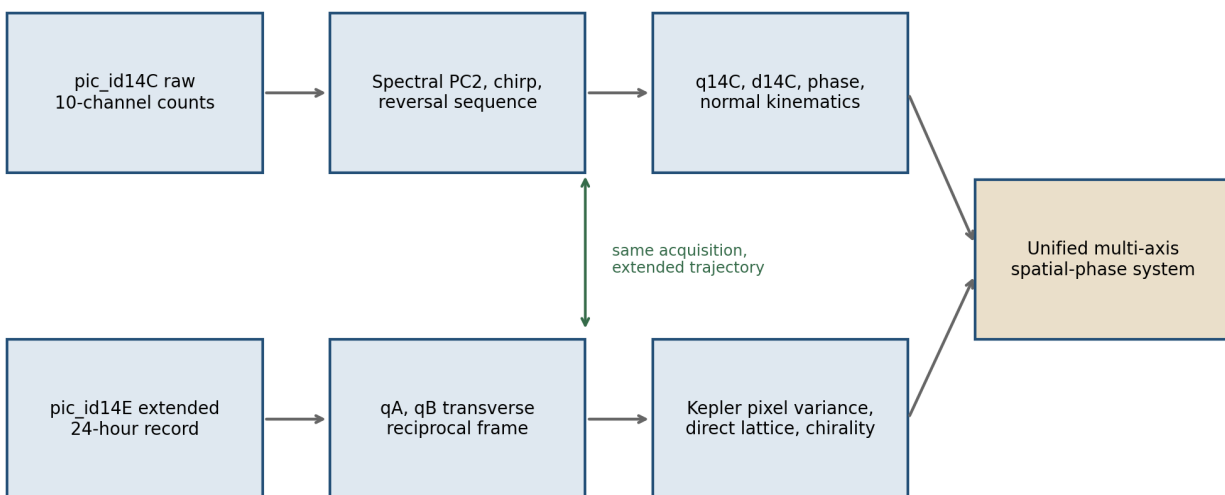


Figure 1. Evidence and derivation graph. The two spectrophotometer analyses are nested records of one acquisition, whereas the Kepler branches provide the independent spacecraft platform. The shared endpoint is a multi-axis spatial-phase representation, not an assertion that every axis or spacing is identical.

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1. Purpose, Scope, and Evidentiary Language

The objective is to provide a self-contained educational reference that fully explains the related findings of the spectrophotometer and Kepler experiments. The report retains the complete directional, kinematic, scale, and signal-state structure developed in the controlling comparison. It also adds the derivations, coordinate transformations, base-cell geometry, event-level cadence analysis, and reproduction details required for independent technical study.

Every major statement is classified by its role. **Observed** quantities are directly present in instrument or event records. **Derived** quantities follow from declared transformations or equations. **Inferred structure** is the minimum geometry that compresses several observed and derived relationships. **Proposed mechanism** describes a sufficient transduction class that can reproduce the observations but is not yet uniquely determined. **Open question** identifies a missing coordinate, measurement, or physical law.

This report does not use statistical significance as a gate for exploratory structure. Numerical fit measures are retained as descriptive facts because they quantify how the proposed coordinate organizes the source signal. The central standard is internal consistency across geometry, trajectory, phase, channel state, scale recurrence, and independent experimental domains.

1.1 Source hierarchy

ID	File	Role	Information used
S1	K-Field_spectrophotometer_pic_id14C_analysis_1785888480.json	pic_id14C archival analysis JSON	Observed metadata, signal decompositions, plane-wave model, channel phases, reversal ledger
S2	pic_id14C_plane_normal_kinematics_time_series.csv	Generated from S1 trajectory and wavevector	Full plane-normal distance, velocity, acceleration, and closure residual time series
S3	pic_id14C_plane_normal_kinematics_summary.json	Summary generated from S2	Terminal and RMS closure values and traversal statistics
S4	K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json	pic_id14E archival exposed-plane JSON	Extended 24-hour reciprocal axes, direct-space base face, nodes, and signal-state ledgers
S5	results_new_translation2_stride_2.txt	Kepler translated-lattice result table	Exact best direct-lattice parameters and 30,000 retained candidates
S6	kepler_lattice_plane_fit_report_v1.pdf	Kepler lattice and line-plane report	Report-defined refined basin and lattice parameter conventions
S7	sheet_lattice_final_paper.pdf	Kepler rank-2 sheet-array report	Sheet basis, dimensions, orientation, and stacking interval
S8	polarity_flips_vs_z_axis.csv	Kepler two-axis event ledger	2,292 chirality reversals, fitted grid-z coordinate, modular coordinates, and state signs
S9	fit_params.json	Kepler 8-fold fit parameters	Grid rotation and D/H periods used for the chirality branch

Table 1. Controlling source hierarchy. All cross-experiment calculations in this report are derived from these records.

1.2 Dependency distinction between pic_id14C and pic_id14E

The pic_id14C record begins at 2026-08-04T10:55:12.483845+00:00 and ends at 2026-08-04T22:53:19.473927+00:00. The pic_id14E source begins at the same effective timestamp and extends to 2026-08-05T10:59:04.362935+00:00. pic_id14C is therefore the first approximately twelve hours of the twenty-four-hour pic_id14E acquisition. The q14C to qA-qB relation is an internal refinement enabled by longer trajectory coverage. It is not an independent-instrument comparison.

The Kepler results supply the independent domain. Kepler uses a spacecraft trajectory, a different detector, pixel-pair variance states, and a 2009 observational interval. The shared geometric relations between Kepler and the spectrophotometer therefore connect distinct platforms and source pipelines.

2. Definitions, Coordinates, and Notation

Symbol	Definition	Units	Role
$R(t)$	Observer position vector	m or km	Earth/CMB/laboratory/device trajectory for spectrophotometer; heliocentric Kepler position for spacecraft
$v(t)$	dR/dt	m/s or km/s	Observer velocity
q	Reciprocal wavevector	cycles/m	Direction and inverse spacing of a plane family
n_{hat}	$q/ q $	dimensionless	Unit normal to equal-phase planes
d	$1/ q $	m or km	Primitive equal-phase spacing
Φ	$2\pi q \cdot R + \phi_0$	rad	Unwrapped spatial phase
f	$q \cdot v$	cycles/s	Plane-crossing frequency
A	$[a_1 a_2 a_3]$ after global rotation	km	Kepler direct-lattice basis matrix
h	Integer reciprocal index	dimensionless	Defines reciprocal family $A^{\wedge T} h$
H_8	8-fold Kepler grid-z period	km	Stored modular period for chirality analysis
H_{sheet}	Rank-2 Kepler sheet stacking interval	km	Third basis length of global sheet array
qA, qB	pic_id14E reciprocal basis vectors	cycles/m	Two independently extracted CMB-transverse directions

Table 2. Core notation and units.

2.1 Heliocentric J2000 ecliptic conversion

The spectrophotometer normals are stored in ICRS/J2000 equatorial Cartesian coordinates. Kepler is modeled in Sun-centered J2000 ecliptic coordinates. A direction vector is converted by a rotation through the J2000 obliquity, $\epsilon = 23.439291111^\circ$:

$$[x_{\text{ecl}}, y_{\text{ecl}}, z_{\text{ecl}}]^{\wedge T} = R_x(-\epsilon) [x_{\text{eq}}, y_{\text{eq}}, z_{\text{eq}}]^{\wedge T}.$$

For directions, velocities, wavevectors, and relative displacements, changing the coordinate origin does not alter orientation or spacing. A Sun-centered origin is required only for absolute position and absolute phase registration. The present comparison therefore establishes common directions and relative spatial phases; it does not assign one common absolute plane index to terrestrial and Kepler events.

2.2 Spatial phase kinematics

A stack of parallel equal-phase planes is represented by one reciprocal wavevector. If q is measured in cycles per metre, the phase is

$$\Phi(t) = 2\pi q \cdot R(t) + \phi_0.$$

Differentiation gives

$$d\Phi/dt = 2\pi q \cdot v(t),$$

and the crossing frequency in cycles per second is

$$f(t) = q \cdot v(t).$$

This identity is the common kinematic core of both experiments. A fixed spacing can produce a changing time period whenever the projected observer velocity changes. Conversely, a fixed sampling cadence can correspond to a changing spatial step.

3. Spectrophotometer Experimental Records

Record	Measurements	Complete turns	Profile/state units	Duration h	Earth rotation deg	Start UTC	End UTC
pic_id14C	84330	937	933	11.96860835611111	180.0206598770981	2026-08-04T10:55:12.483845+00:00	2026-08-04T22:53:19.473927+00:00
pic_id14E	169555	1883	multiple full-valid apertures	24.064410858333336	361.954456	2026-08-04T10:55:12.483845+00:00	2026-08-05T10:59:04.362935+00:00

Table 3. Spectrophotometer source records. The extended record contains the shorter record as its first interval.

The detector direction is reconstructed at every sample in the inertial frame. The trajectory includes the supplied CMB translation, the rotating Earth, the laboratory position, and the detector offset on the rotating instrument. Repeated mechanical orientations are therefore different inertial positions rather than repeated samples of one closed local circle.

$$R(t, \theta) = g(t - t_0) + r_{\text{lab, ICRS}}(t) + r_{\text{device}} u_{\text{ICRS}}(t, \theta).$$

4. The pic_id14C Multichannel State and One-Axis Solution

4.1 Low-dimensional spectral organization

Band (turn periods)	PC1 fraction	PC1+PC2 cumulative	Participation dimension
short_3_12	0.782511	0.845801	1.610565
mid_12_64	0.657715	0.829468	2.137104
slow_64_300	0.551853	0.899915	2.331966

Table 4. Effective dimensionality of the ten-channel state. The shortest band is strongly common-mode; slower bands expose additional spectral structure.

The 12-64-turn band contains the clearest non-common mode. PC1 carries 65.7715 percent of standardized variance and is broadband. PC2 carries 17.1754 percent of total variance and 50.1785 percent of the variance remaining after PC1. Its loadings divide the channels into a blue pole and an opposite lower-energy/NIR pole.

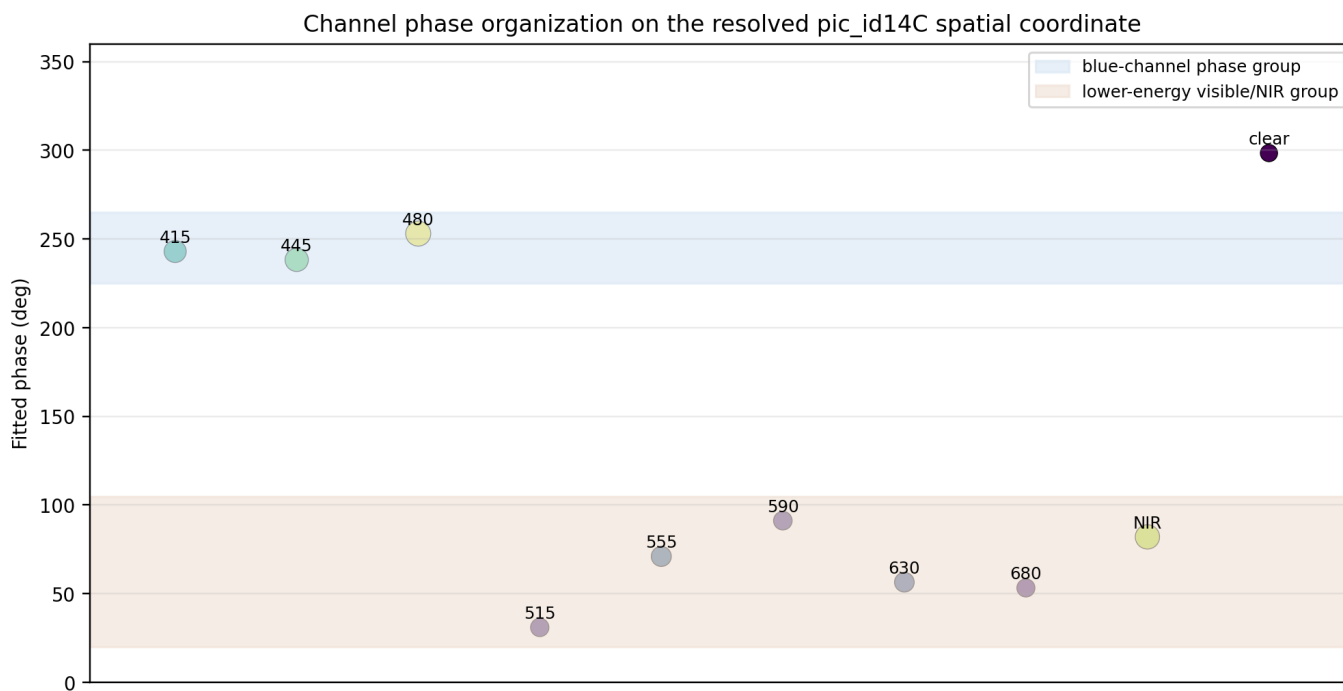


Figure 2. Channel phases on the common spatial coordinate. Marker area and color encode the channel plane-wave R-squared. The 415, 445, and 480 nm channels occupy one compact phase group; the lower-energy visible channels and NIR occupy the nearly antipodal group.

4.2 Resolved wavevector and phase landmarks

Quantity	Value	Units	Status
Wavevector qx	1.913377581942e-07	cycles/m	Fitted
Wavevector qy	-4.800188656215e-07	cycles/m	Fitted
Wavevector qz	3.750569111340e-07	cycles/m	Fitted
Normal ICRS	(0.299662563394, -0.751778870555, 0.587393290639)	dimensionless	Derived from q
RA / Dec	291.732456586 / +35.972245579	deg	Derived direction
Ecliptic normal	(0.299662563394, -0.456091995814, 0.837963268559)	dimensionless	Coordinate conversion
Ecliptic lon / lat	303.305802120 / +56.925666655	deg	Coordinate conversion
Primitive spacing	1,566.144426	km	1/ q
Sign-reversal spacing	783.072213	km	d/2
Three-quarter phase	1,174.608319	km	3d/4
Angle to supplied CMB axis	89.946853887	deg	Axial angle

Table 5. Complete pic_id14C plane-wave coordinate.

Dimensionally correct primitive interval and phase landmarks

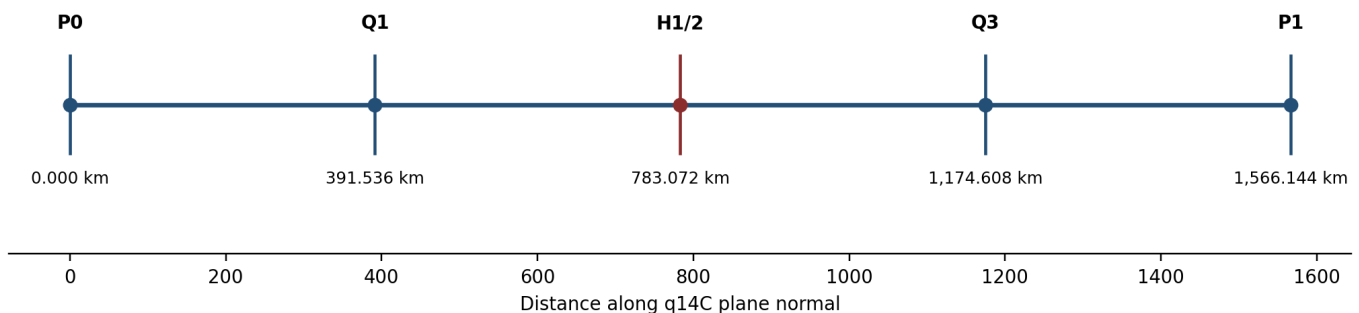


Figure 3. Dimensionally correct primitive interval. P0 and P1 are equal-phase planes separated by 1,566.144 km. H1/2 is the spectral sign-reversal sheet. Q1 and Q3 are quarter-cycle landmarks.

4.3 Measured chirp and projected velocity

The observed period contracts from 44.522 to 29.574 turns. The plane model predicts a contraction from 43.434 to 30.023 turns. The resolved normal is almost perpendicular to the supplied CMB direction. The constant CMB contribution projects to 863.410 m/s, while Earth rotation contributes from -79.288 to 270.981 m/s. Because the dominant translation is nearly tangent to the planes, the smaller rotating contribution substantially changes the total normal speed.

The extracted frequency track and the projected velocity coordinate have R-squared 0.949759. The fitted phase organizes PC2 with R-squared 0.658970, organizes the blue-red/NIR contrast with R-squared 0.631155, and retains held-out phase R-squared 0.601514. Twenty-four of twenty-nine fitted reversal events are matched within five rotations.

These values are descriptive measured-to-model relationships. Their importance is that frequency, continuous spectral state, and discrete reversals all refer to the same recovered spatial coordinate.

Channel	Plane-wave R2	Amplitude	Phase deg	Shift-null percentile
415	0.372062	0.740841	242.975	100.000
445	0.457417	0.652038	238.230	100.000
480	0.601312	1.251609	253.054	100.000
515	0.123141	0.173917	30.955	97.656

Channel	Plane-wave R2	Amplitude	Phase deg	Shift-null percentile
555	0.230059	0.387285	71.003	100.000
590	0.138933	0.193531	91.164	98.438
630	0.206067	0.335646	56.471	98.438
680	0.107285	0.163760	53.226	88.281
NIR	0.540354	0.904296	82.014	100.000
clear	0.059378	0.197769	298.433	67.969

Table 6. Channel-resolved response to the common spatial phase. The blue and NIR channels carry the strongest individual responses; lower-energy channels remain phase organized even where individual R-squared is smaller.

5. Plane-Normal Distance, Velocity, Acceleration, and Closure

5.1 Distance derived from phase

One full phase cycle corresponds to one primitive spacing. Relative displacement is therefore

$$x_{\text{phase}}(t) = d_{14C} [\Phi_{\text{cycles}}(t) - \Phi_{\text{cycles}}(t_0)].$$

The full record advances 38,592.740 km along the normal, equal to 24.641878 primitive spacings and 49.283756 half-spacing reversal intervals.

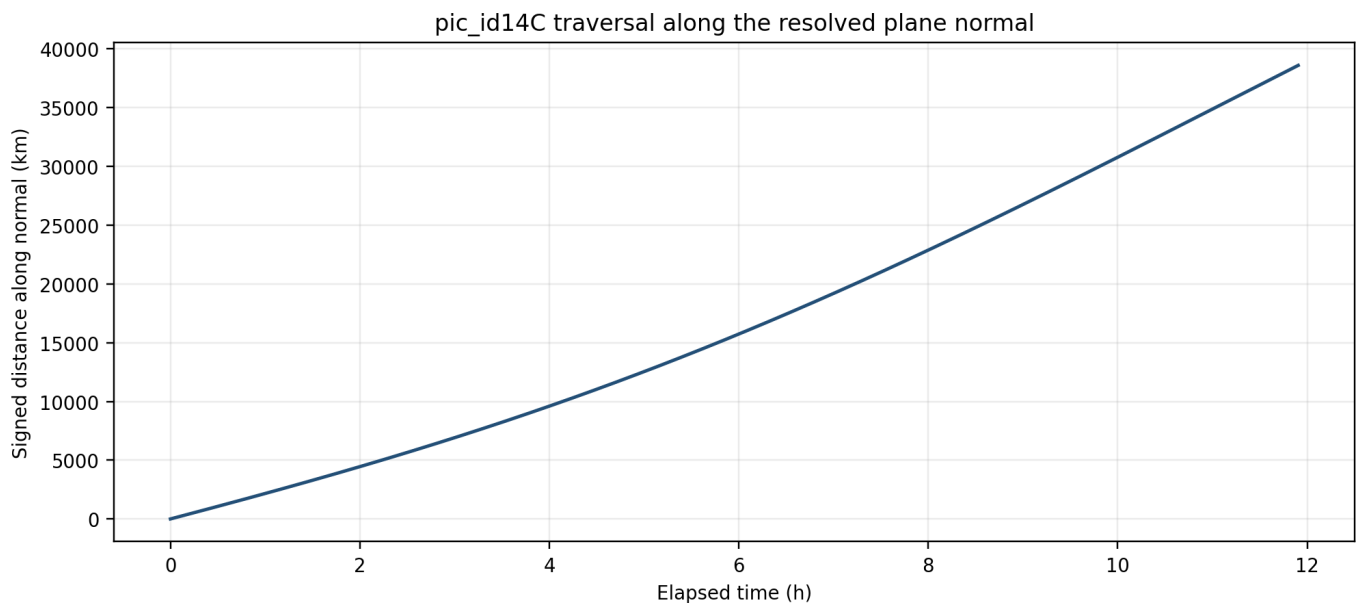


Figure 4. Signed displacement along the resolved plane normal. The path is monotonic in this interval because the projected normal velocity remains positive.

5.2 Velocity and acceleration

Normal velocity is the trajectory projection

$$v_n(t) = \hat{n}_{14C} \cdot v(t).$$

Acceleration is the time derivative of that projection. It is a kinematic acceleration of the observer coordinate, not a direct force measurement.

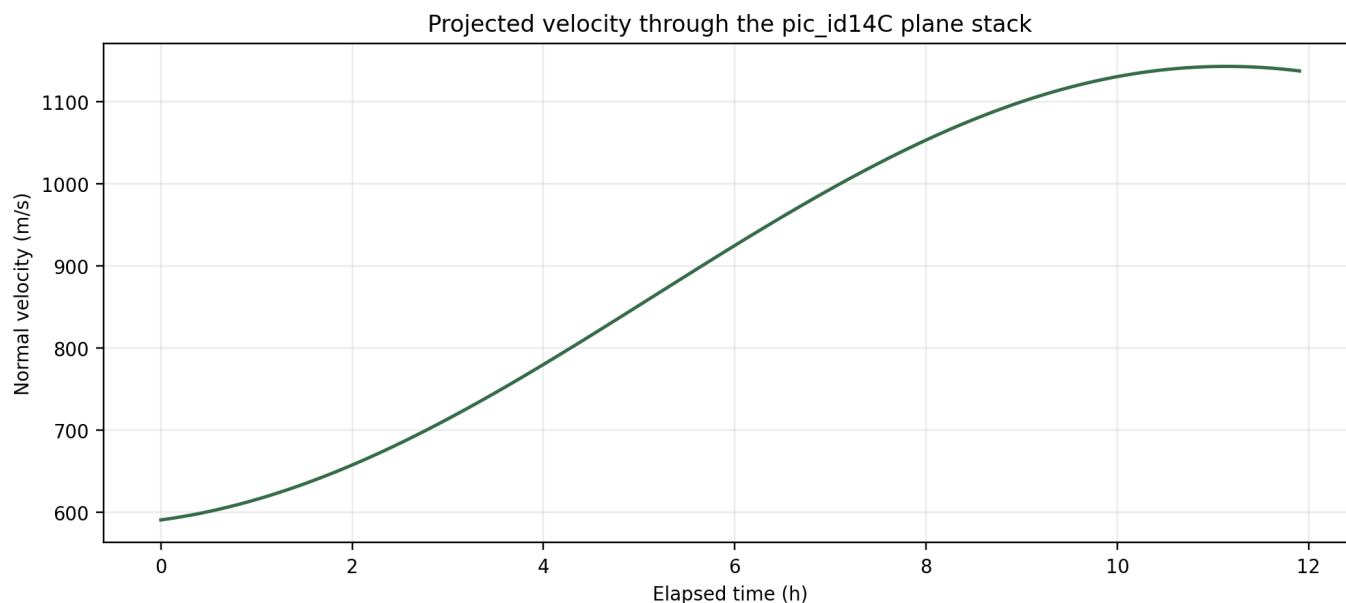


Figure 5. Normal velocity increases from approximately 590.375 m/s to 1,142.998 m/s across the complete profile sequence.

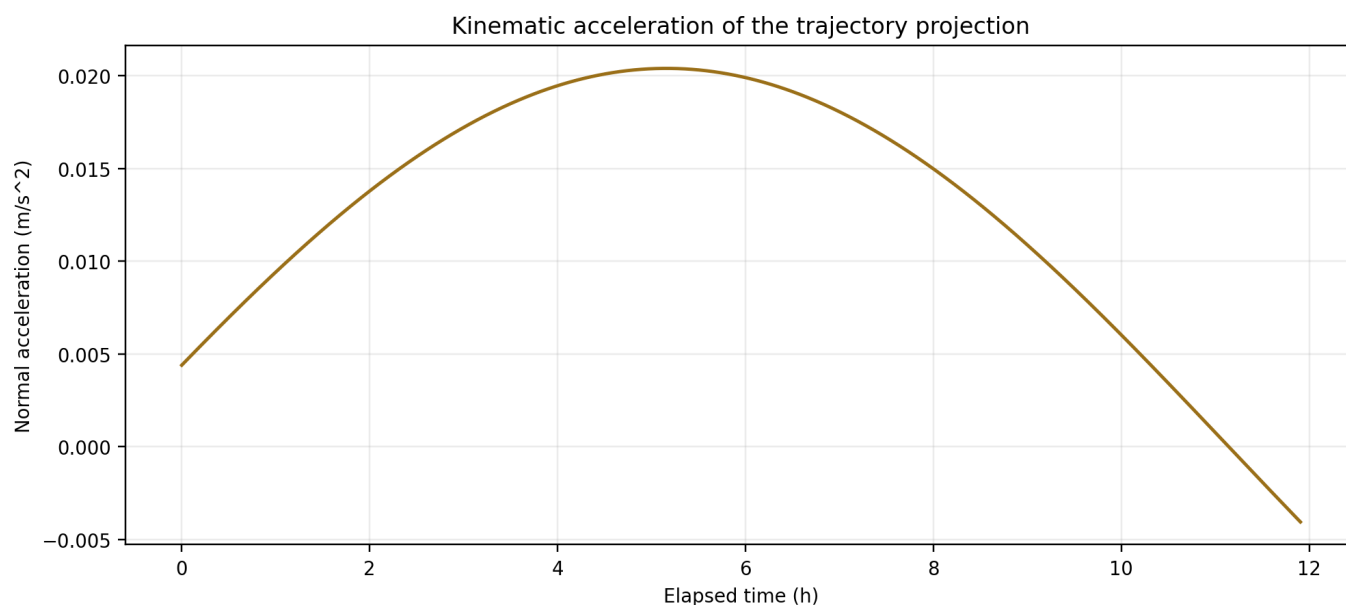


Figure 6. Plane-normal acceleration. The smooth positive-to-negative curvature follows the changing projection of Earth rotation on the fixed normal.

Quantity	Value	Units	Construction
Signed distance	38,592.739608	km	Derived from unwrapped phase
Primitive spacings traversed	24.641877836	cycles	Distance / d14C
Velocity minimum	590.374805	m/s	Trajectory projection
Velocity maximum	1142.997631	m/s	Trajectory projection
Velocity mean	900.448176	m/s	Time-series mean
Acceleration minimum	-0.004061755	m/s ²	Numerical derivative
Acceleration maximum	0.020387835	m/s ²	Numerical derivative
Acceleration RMS	0.014443756	m/s ²	RMS over record

Table 7. pic_id14C plane-normal kinematics.

5.3 Independent velocity integration

The second distance route integrates the projected velocity. For nonuniform sample intervals, the trapezoidal rule is

$$x_v(t_k) = \sum_i 0.5 [v_n(i) + v_n(i+1)] [t_{i+1} - t_i].$$

At the final sample, x_{phase} and x_v differ by 1.681951 m. Their RMS difference over the full time series is 2.148123 m. The terminal relative discrepancy is 4.358204700e-08, or 0.043582 ppm.

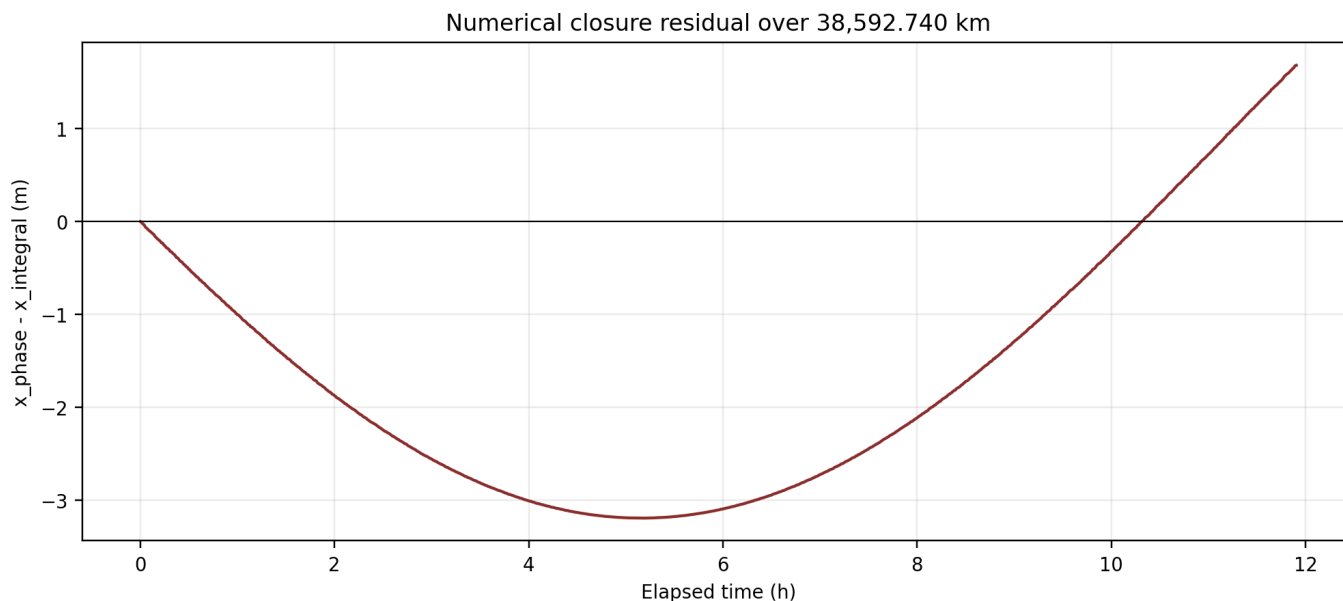


Figure 7. Difference between phase-derived distance and trapezoidally integrated normal velocity. The residual remains metre-scale over a 38.59-million-metre traversal.

This closure verifies that the fitted phase, spacing, wavevector direction, timestamps, reconstructed trajectory, and projected velocity implement one consistent spatial coordinate. The measured signal enters upstream by constraining the chirp, spectral state, and reversal sequence. The closure is therefore the terminal mathematical agreement of a coordinate derived from measured modulation.

6. The pic_id14E CMB-Transverse Reciprocal Frame

6.1 Independently extracted reciprocal directions

Axis	Originating state	Spacing km	RA deg	Dec deg	Ecl lon	Ecl lat	Angle to CMB
qA	qA_spectral_PC2	317.864281	81.380625	+45.795586	83.503479	+22.550486	89.984183
qB	qB_std31_angular_PC1	1,034.800487	296.484362	+44.633516	316.144962	+63.890323	89.965843
qC	qC_broadband_PC1	260.276750	46.645302	+69.405455	68.306409	+49.210158	89.996828
qF	bridge_carrier_qF	670.885616	79.653162	+48.332024	82.419106	+25.164657	89.983619

Table 8. Representative pic_id14E reciprocal directions. qA and qB originate in different observables and define the principal transverse frame.

qA and qB are separated by 84.391343 deg. Their normalized cross product is 0.036386 deg from the supplied CMB axis. The relation is directional closure in three dimensions: two independently extracted reciprocal coordinates reconstruct the dominant translation line.

$$c_{\text{hat}} = (qA \times qB) / |qA \times qB|.$$

The longer trajectory therefore changes the model from one isolated plane family to a two-axis reciprocal frame embedded transverse to the dominant translation direction.

6.2 Dimensionally recovered direct-space base face

The minimum-norm direct vectors $\mathbf{a}$ and $\mathbf{b}$ are defined by the reciprocal duality conditions

$$\mathbf{qA} \cdot \mathbf{a} = 1, \mathbf{qB} \cdot \mathbf{a} = 0, \mathbf{qA} \cdot \mathbf{b} = 0, \mathbf{qB} \cdot \mathbf{b} = 1.$$

The resulting direct vectors have lengths 319.393331 km and 1039.778275 km. Their included angle is 95.608657 deg and their area is 330,508.374179 km². The reciprocal plane spacings are 317.864281 km and 1034.800487 km.

pic_id14E exposed direct-space base face and complete 4 x 4 node patch

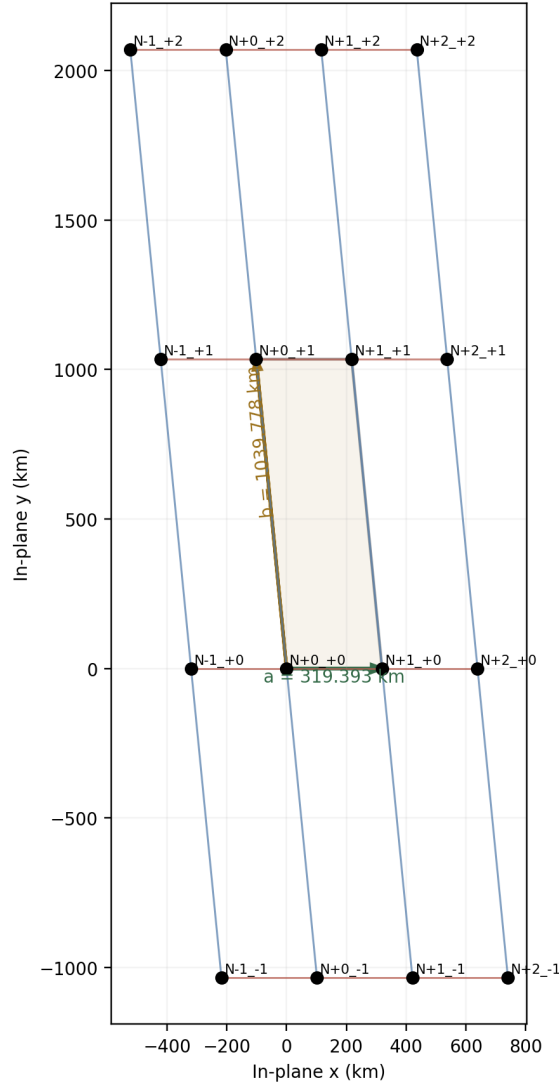


Figure 8. Dimensionally correct orthonormal projection of the exposed base face. All 16 nodes in the stored display patch are labeled. Blue lines are constant $\mathbf{qA}$ phase, red lines are constant $\mathbf{qB}$ phase, and the shaded parallelogram is the primitive face $\mathbf{N00-N10-N11-N01}$.

Dimensionally correct pic_id14E base-face patch and 500 km direction arrows

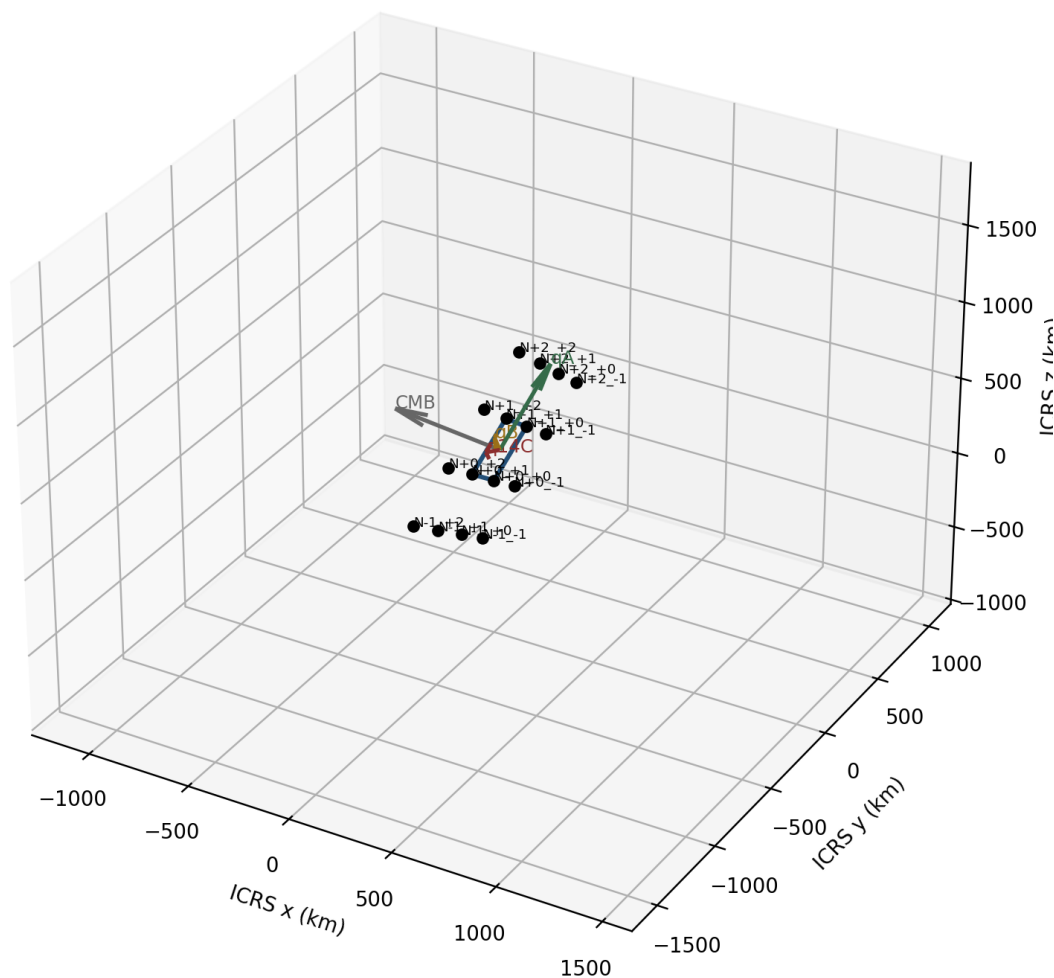


Figure 9. Three-dimensional ICRS rendering of the same node patch. Direction arrows are exactly 500 km long and show qA, qB, q14C, and the supplied CMB direction. The figure preserves physical coordinate scales.

Node	i	j	x km	y km	z km	qA cycles	qB cycles
N00	0	0	0.000000	0.000000	0.000000	0.000000	0.000000
N10	1	0	23.580892	241.209359	208.024245	1.000000	0.000000
N01	0	1	320.888448	-735.846440	660.832416	-0.000000	1.000000
N11	1	1	344.469340	-494.637081	868.856661	1.000000	1.000000

Table 9. Four primitive base-face nodes and exact phase coordinates.

6.3 Scope of the exposed geometry

The qA-qB pair defines a complete transverse base face, but it does not determine the longitudinal direct-space period. The dominant translation is approximately normal to the qA-qB reciprocal plane, so the trajectory provides strong transverse phase information and weak independent leverage on a third reciprocal coordinate. The measured object is therefore a two-dimensional base face embedded in three-dimensional motion, not a closed three-dimensional primitive cell.

Additional pic_id14E axes qC and qF occupy recurring transverse sectors and organize broadband and carrier states. They demonstrate multiplex organization across scale and signal construction. They should not be collapsed into one universal spacing.

7. Internal Spectrophotometer Bridge: q14C Within the qA-qB Frame

The key internal refinement is obtained by solving the least-squares vector equation

$$q14C = \alpha qA + \beta qB + r.$$

The fitted coefficients are $\alpha = -0.0332521971$ and $\beta = 0.6624680143$. The residual magnitude is 0.148007 percent of $|q14C|$, and q14C lies only 0.084802 deg from the qA-qB plane.

$$q14C = -0.0332521971 qA + 0.6624680143 qB + r.$$

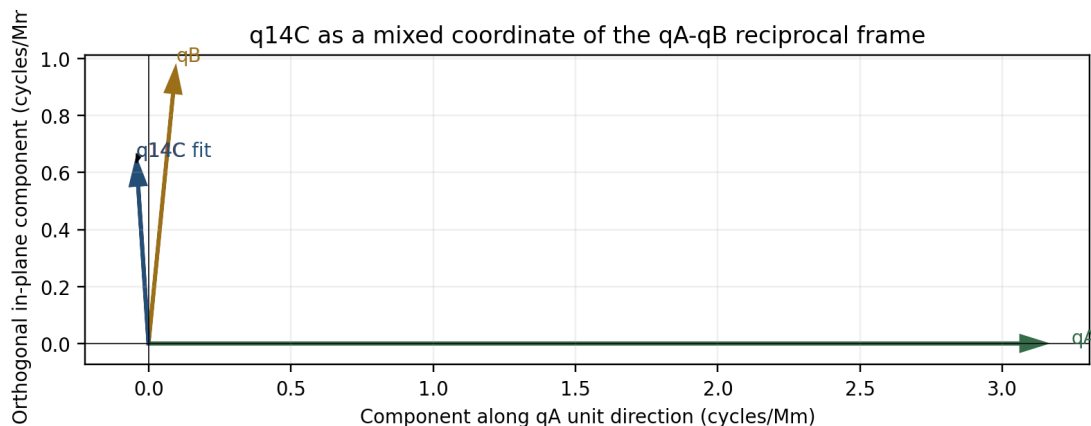


Figure 10. Reciprocal-space decomposition of q14C in an orthonormal basis spanning qA and qB. The residual arrow is strongly magnified by the plot scale; its magnitude is 0.1480 percent of $|q14C|$.

A compact exploratory rational approximation is

$$q14C \text{ approximately equals } -(1/30) qA + (2/3) qB.$$

This rational approximation has 0.6499 percent vector mismatch and 0.0922 deg axial angular mismatch. The direct least-squares coefficients remain the authoritative decomposition; the rational form is retained as a candidate arithmetic grammar.

The structural consequence is that the one-axis pic_id14C result is not isolated. It is a mixed reciprocal coordinate exposed before the longer trajectory resolved the two basis directions separately.

Item	Value	Meaning	Status
Least-squares alpha	-0.0332521971	Coefficient of qA	Derived
Least-squares beta	+0.6624680143	Coefficient of qB	Derived
Relative vector residual	0.0014800701	0.148007 percent	Derived
Angle from qA-qB plane	0.084801800	deg	Derived
Rational alpha	-1/30	Candidate reduction	Inferred arithmetic
Rational beta	2/3	Candidate reduction	Inferred arithmetic
Rational relative mismatch	0.006498787	0.6499 percent	Derived

Table 10. Complete q14C decomposition ledger.

8. Kepler Measurement Constructions

8.1 Pixel-variance observables

The Kepler analysis uses custom secondary observables built from pixel-level data. For the single scalar branch, adjacent-pixel contrast is converted to a rolling standard deviation named moving_std_diff. For the two-axis branch, two approximately orthogonal pixel-pair differentials produce X and Y moving-standard-deviation signals.

The two-axis state is represented as a complex number

$$z(t) = X(t) + i Y(t).$$

Its unwrapped angle and angular velocity are

$$\theta(t) = \text{unwrap}[\arg z(t)], \omega(t) = d\theta/dt.$$

A chirality reversal is recorded when the sign of ω changes. The new sign is stored in the event ledger. The analysis aperture is 31 Kepler samples, so the observed state includes the effect of a declared 31-sample variance window.

8.2 Independent geometric branches

Branch	Signal input	Scale	Role
Translated skew direct lattice	moving_std_diff scalar signal	L approximately 3.74-3.78 million km	Contains reciprocal family (1,-2,-3) aligned with q14C
Rank-2 sheet array	collapse of frequency-component lattice nodes	L1=2,258,266 km; L2=2,244,300 km; H=655,343 km	Global sheet and stacking geometry
8-fold two-axis grid	X/Y variance and chirality	D=1,550,000 km; H8=228,650 km	Grid-z coordinate for 2,292 reversal events
Ecliptic XY projection	projected reversal/FFT studies	axis near 98 deg; spacing near 1,795 km	Separate in-plane branch

Table 11. Kepler geometric branches. They are coupled by the source corpus but are not the same axis or spacing.

9. Kepler Translated Direct Lattice

9.1 Direct basis

The direct basis is constructed from one scale and two internal skew angles:

$$a_1 = L [1, 0, 0]^T,$$

$$a_2 = L [\cos(\alpha), \sin(\alpha), 0]^T,$$

$$a_3 = L [\cos(\beta), \sqrt{1-\cos(\beta)^2} \cos(\alpha), \sqrt{1-\cos(\beta)^2} \sin(\alpha)]^T.$$

The physical basis is

$$A = R_z(\text{yaw}) R_y(\text{tilt}) [a_1 \ a_2 \ a_3].$$

The translation coordinates q_1 , q_2 , and q_3 locate the lattice origin within the direct cell. Reciprocal directions are generated from A^{-T} and do not depend on the translation phase.

Parameter	Exact best row	Refined basin
score	0.391457800	0.388345100
L_km	3781125.000000000	3739968.360000000
yaw_deg	-8.936985000	-12.033914000
tilt_deg	-143.867800000	-144.874256000
skew1_deg	78.316060000	76.224157000
skew2_deg	-43.652510000	-41.109722000
q1	0.414155000	0.402297800
q2	0.514634800	0.505406800
q3	0.652949300	0.644155100

Table 12. Kepler direct-lattice parameter sets used in the comparison.

Exact Kepler direct cell and reciprocal plane (1,-2,-3)

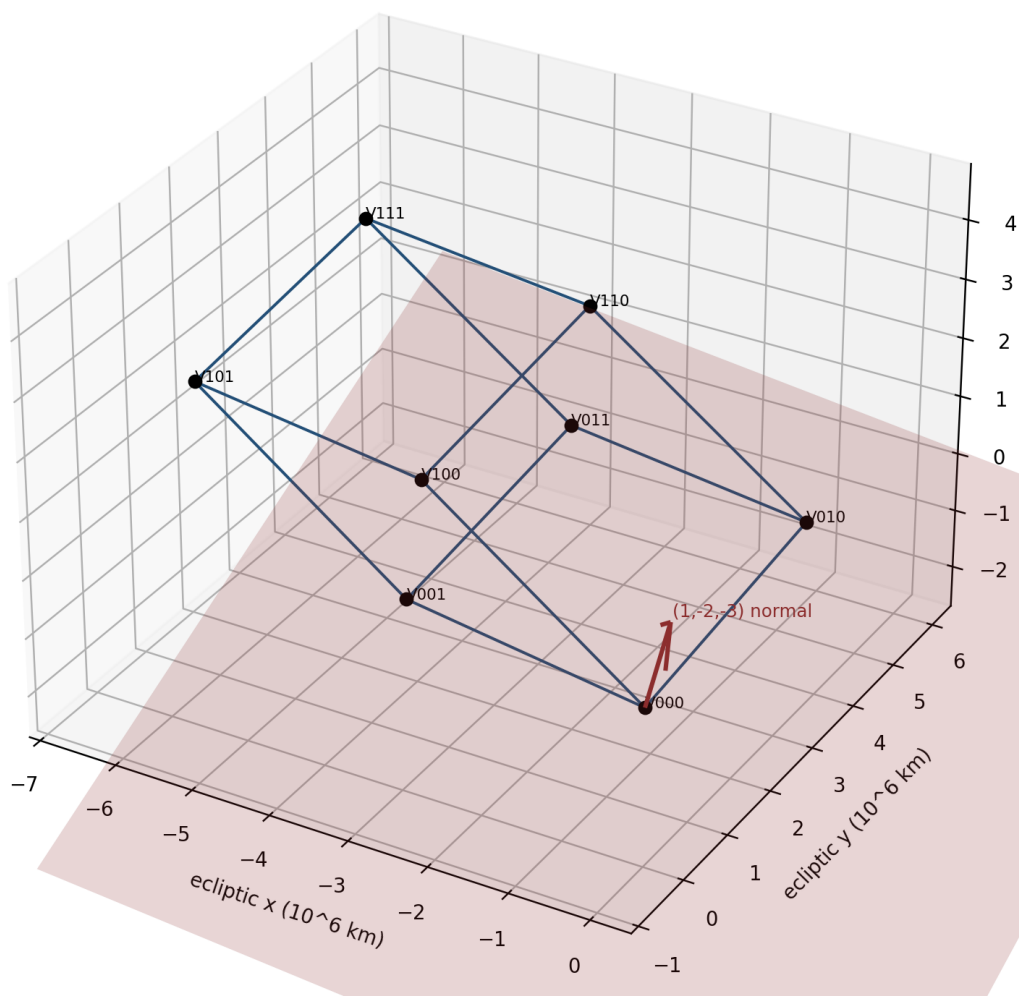


Figure 11. Exact Kepler direct cell. All eight corner nodes are labeled in heliocentric J2000 ecliptic coordinates. The translucent plane is the reciprocal family indexed by (1,-2,-3) through the origin.

Refined Kepler direct cell and reciprocal plane (1,-2,-3)

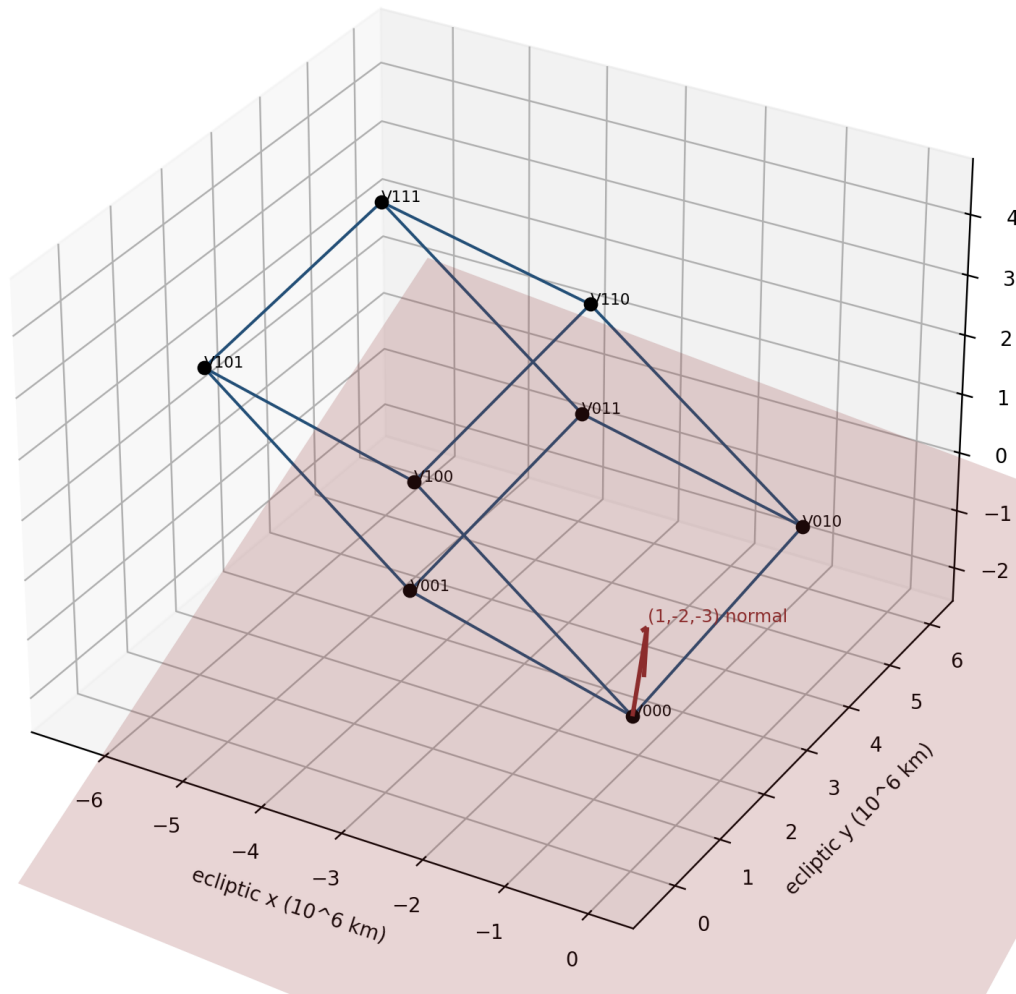


Figure 12. Refined Kepler direct cell and the same low-order reciprocal family. The refined normal provides the closest directional match to q14C.

10. Common Reciprocal Direction: Kepler (1,-2,-3) and q14C

10.1 Reciprocal-plane construction

For integer reciprocal index $h = (1,-2,-3)$, the reciprocal vector, unit normal, and interplanar spacing are

$$g_h = A^{-T} h, n_h = g_h / |g_h|, d_h = 1 / |g_h|.$$

Solution	J2000 ecliptic unit normal	Spacing km	Angle to q14C deg	Angle to qA-qB plane deg
Exact Kepler	(0.319039300, -0.455876340, 0.830897519)	673,775.299	1.181800	1.096651
Refined Kepler	(0.290854524, -0.462941572, 0.837310425)	636,407.838	0.640396	0.560784
Exact/refined midpoint	(0.304980669, -0.459459811, 0.834196304)	not assigned	0.420313	not assigned

Table 13. Common-direction comparison in one coordinate frame.

The exact Kepler normal is 1.181800 deg from q14C. The refined normal is 0.640396 deg from q14C. The normalized midpoint of the two Kepler normals is 0.420313 deg from q14C. Both Kepler solutions also lie close to the qA-qB plane exposed by pic_id14E.

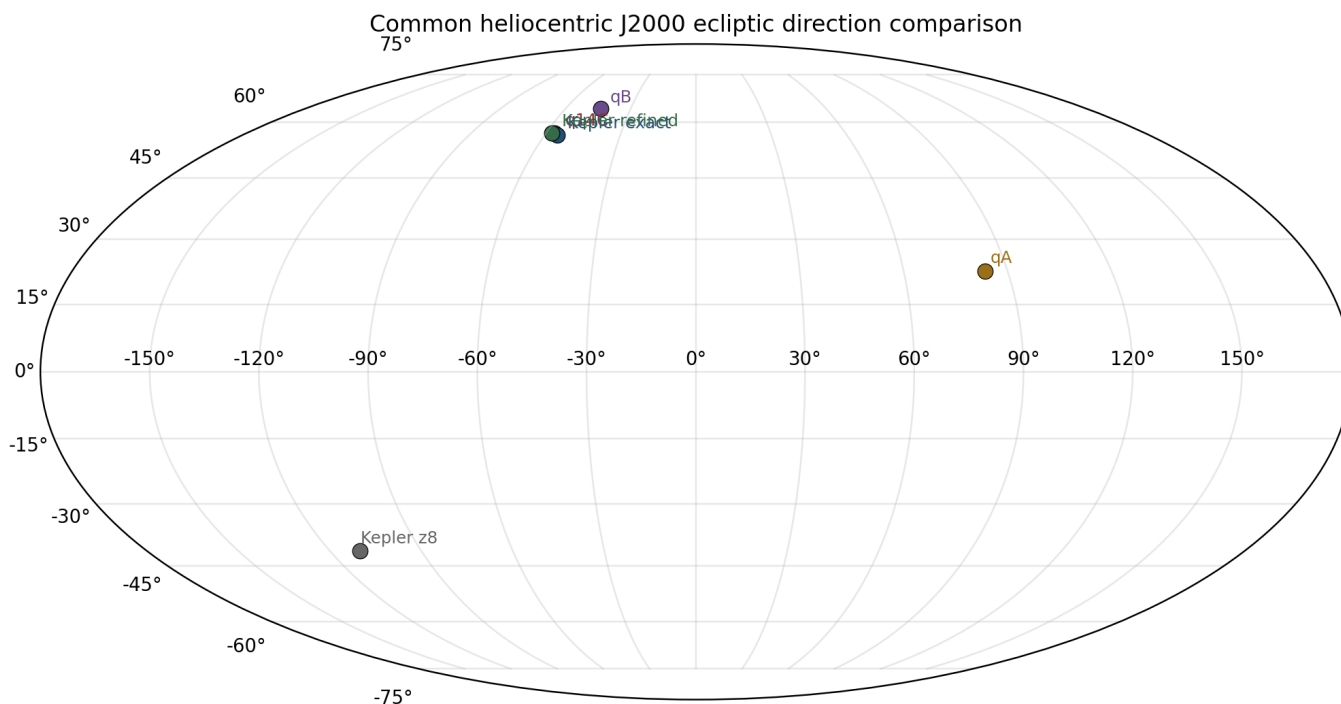


Figure 13. Heliocentric J2000 ecliptic direction map. q14C, the exact and refined Kepler (1,-2,-3) normals, qA, qB, and the independent Kepler 8-fold z-axis are displayed in one frame.

10.2 Complete plane-subspace relation

Direct integer vector	Exact angle from q14C plane deg	Refined angle deg	h dot (u,v,w)
(1, 2, -1)	0.104076	0.402208	0
(2, 1, 0)	0.951108	0.094509	0
(3, 0, 1)	1.154259	0.335254	0

Table 14. Low-order Kepler direct lines constrained to the (1,-2,-3) reciprocal plane. $h \cdot (u,v,w)=0$ confirms exact membership in the Kepler plane; the angular columns compare the realized lines with the spectrophotometer plane.

The common relation is therefore not limited to a single normal-vector comparison. A low-order integer subspace of the Kepler direct lattice lies within the spectrophotometer plane. This supplies both a normal alignment and a direct-line containment relationship.

10.3 Relation to the pic_id14E basis

When the Kepler normals are converted back into the ICRS basis and decomposed in the qA-qB unit-vector plane, the exact solution is approximately $-0.1643 \text{ nA} + 1.0024 \text{ nB}$, and the refined solution is approximately $-0.1712 \text{ nA} + 1.0021 \text{ nB}$. Their residual angles from the qA-qB plane are 1.096651 deg and 0.560784 deg, respectively.

The Kepler reciprocal family and q14C therefore occupy the same mixed-direction sector of the two-axis spectrophotometer frame.

11. Kepler Rank-2 Sheet Array

The global Kepler sheet model is a repeated array rather than one translated direct cell:

$$r(n_1, n_2, n_3) = r_s + n_1 b_1 + n_2 b_2 + n_3 b_3.$$

Before rotation, b_1 and b_2 span the sheet and b_3 is the transverse stacking vector:

$$B_0 = [[L_1, L_2 \cos(\gamma), 0], [0, L_2 \sin(\gamma), 0], [0, 0, H]].$$

$$B = R_z(\text{yaw}) R_y(\text{tilt}) R_x(\text{roll}) B_0, r_s = B q_s.$$

Sheet parameter	Value
L1_km	2258266.0
L2_km	2244300.0
H_km	655343.0
gamma_deg	22.4424
yaw_deg	-87.731
tilt_deg	-119.3409
roll_deg	29.5879
q	[0.468, 0.7128, 0.635]

Table 15. Stored Kepler rank-2 sheet-array parameters.

Kepler rank-2 sheet array: complete 3 x 3 x 3 node neighborhood

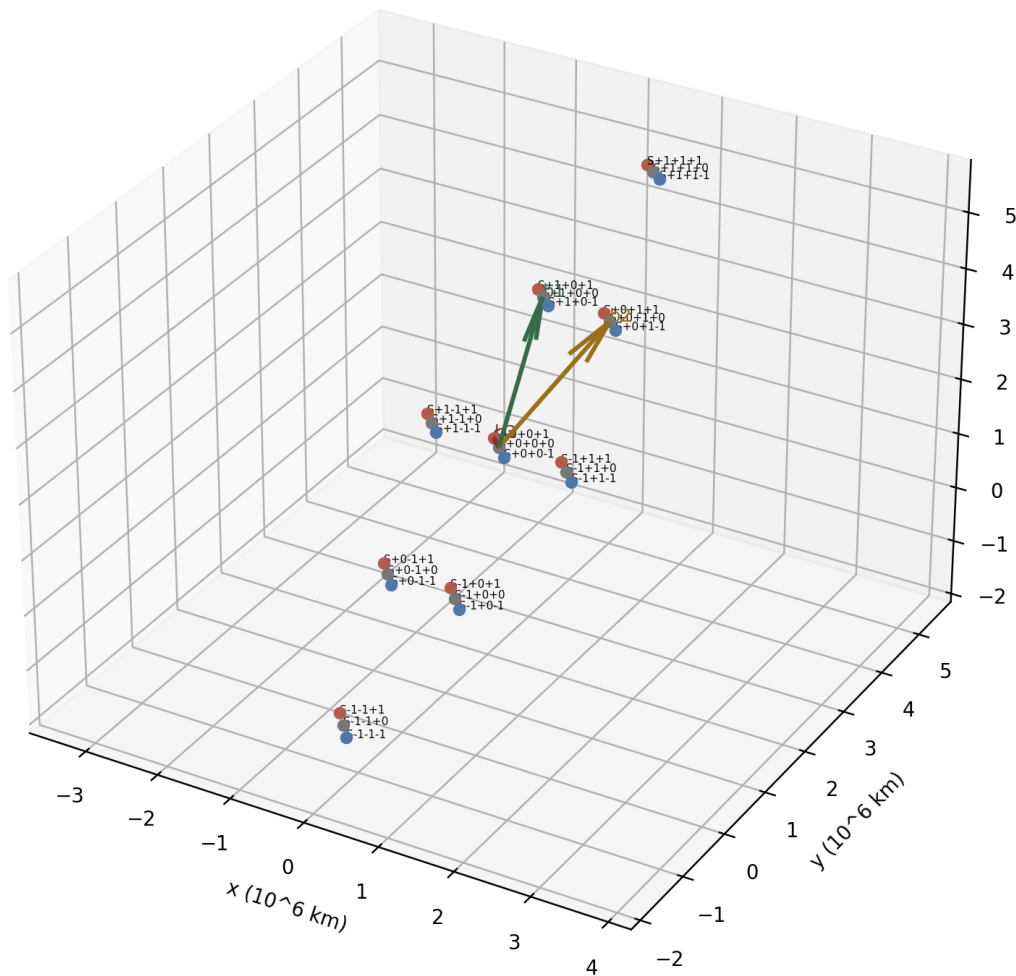


Figure 14. Complete 3 x 3 x 3 neighborhood of the fitted Kepler sheet array. All 27 nodes are labeled. The two million-kilometre basis vectors span each sheet; b3 advances 655,343 km between sheets.

The exact and refined Kepler (1,-2,-3) spacings are 673,775.299 km and 636,407.838 km. Their midpoint is 655,091.569 km, only 251.431 km or 0.03837 percent from the independent sheet interval $H = 655,343.000$ km.

This is a Kepler-internal scale recurrence. The printed sheet stacking direction b3 is not the q14C direction; scalar spacing convergence and directional convergence belong to different branches and must remain separate.

12. Kepler Two-Axis Chirality and Event-Level Traversal

12.1 Event ledger facts

Item	Value	Units	Status
Events	2292	rows	Observed
BJD start	260.252816511533	BJD	Observed
BJD end	289.960259341020	BJD	Observed
Duration	29.707442829	days	Derived
z start	26,881,291.409894	km	Observed
z end	-24,860,159.695302	km	Observed
Signed z travel	-51,741,451.105196	km	Derived
Mean projected speed	20.158564008	km/s	Derived
Positive/negative states	1146/1146	events	Observed
Strict alternation	True	boolean	Observed

Table 16. Kepler polarity-reversal ledger summary.

12.2 Exact cadence decomposition

Every event interval is an integer multiple of the minimum cadence $\Delta t_0 = 58.845500946$ s. The maximum difference between an observed interval and $N \Delta t_0$ is $5.613575922e-06$ s.

$$\Delta t_i = N_i \Delta t_0, N_i \text{ in positive integers.}$$

The integer run lengths range from 1 to 1143. The first four run lengths contain 850 of 2291 intervals. The 31-34 cadence family contains 182 intervals and occurs at the same scale as the declared 31-sample moving-standard-deviation aperture.

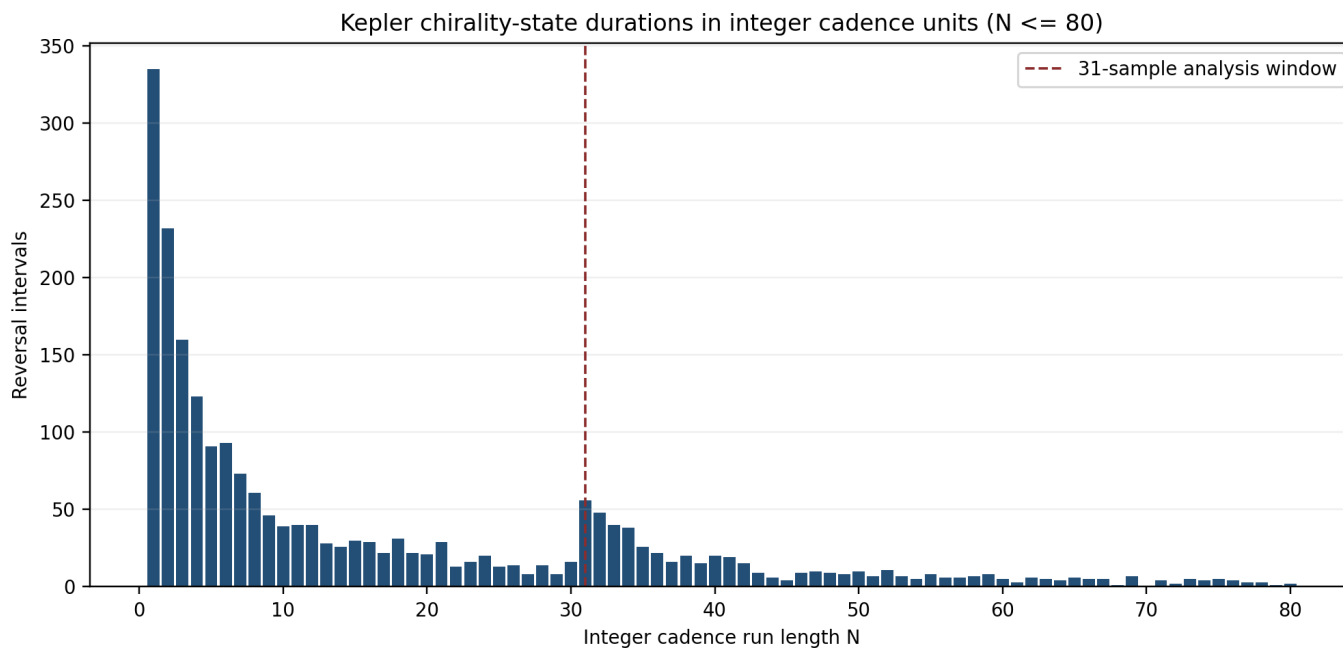


Figure 16. Distribution of Kepler chirality-state durations for $N \leq 80$. The secondary concentration near $N=31-34$ is shown relative to the 31-sample analysis window.

Cadence run N	Intervals
1	335
2	232
3	160
4	123
5	91
6	93
7	73
8	61
9	46
10	39
31	56
32	48
33	40
34	38

Table 17. Selected Kepler cadence-run counts.

12.3 Spatial distance per cadence

For each reversal interval, the signed grid displacement is Δz_i . The spatial distance can be factored into integer state duration and distance per spacecraft cadence:

$$|\Delta z_i| = N_i s(t_i), s(t_i) = v_z(t_i) \Delta t_0.$$

The one-cadence distance decreases smoothly from 1,297.704 km to 1,018.836 km as the projected spacecraft speed falls from 22.052724 to 17.313738 km/s. Its mean is 1,178.235 km and its median is 1,186.712 km.

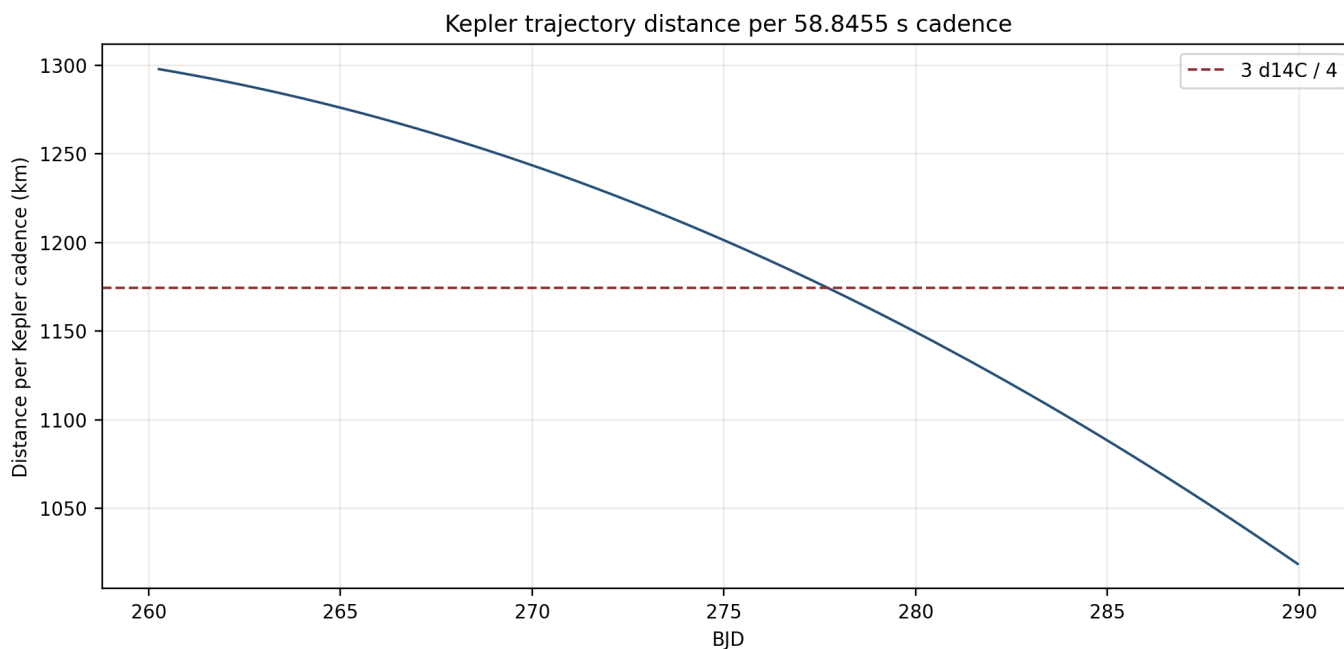


Figure 15. Distance traveled along the Kepler fitted grid-z axis during one 58.8455 s cadence. The curve is velocity-controlled. The dashed line marks the spectrophotometer three-quarter-cycle distance.

The earlier 1,200-1,300 km reversal band is therefore the early one-cadence branch of a changing trajectory function. Higher bands are generated by 2s(t), 3s(t), 4s(t), and larger integer state durations. The low-occupancy intervals between bands arise from discrete N combined with a smooth velocity sweep.

13. Kepler 8-Fold Axis and Other Directional Branches

The grid-z axis associated with the 8-fold chirality fit is $n_8 = (-0.258905088, -0.705252783, -0.659989899)$ in J2000 ecliptic coordinates, with longitude 249.841328 deg and latitude -41.299102 deg. Its axial separation from q14C is 72.002752 deg. It is therefore not the common Kepler reciprocal family identified in Chapter 10.

After conversion to ICRS, the same axis is 7.984031 deg from the pic_id14E broadband direction qC. This is a secondary relation connecting the Kepler chirality branch to a different spectrophotometer state coordinate.

The approximately 98 deg ecliptic projection axis and approximately 1,795 km projected spacing are another distinct branch. The current corpus therefore contains several coupled directions. The correct combined representation is multi-axis.

Axis A	Axis or plane B	Axial angle deg	Interpretation
Kepler reciprocal (1,-2,-3)	q14C	0.640396	Primary common-direction relation
Kepler 8-fold grid z	q14C	72.002752	Distinct axis
Kepler 8-fold grid z	pic_id14E qC	7.984031	Secondary state-mode relation
q14C	qA-qB plane	0.084802	Internal spectrophotometer mixed mode

Table 18. Directional relations that must remain distinct.

14. Cross-Scale Relations

14.1 Fixed H8 to d14C commensurability

The stored 8-fold period is $H_8 = 228,650.000$ km. Relative to d14C = 1,566.144426 km,

$$H_8 / d14C = 145.995475429.$$

Thus H8 is close to 146 d14C. The explicit difference $H_8 - 146 d14C$ is -7.086131 km, equal to approximately -30.991 ppm. The half-period relation is $H_8/2 = 72.997737715 d14C$, close to 73 d14C.

H8 is stored to limited precision in the fit-parameter record. The result is therefore a near-integer commensurability, not a sub-kilometre equality claim.

14.2 Dynamic cadence-to-quarter-phase relation

The mean Kepler distance per cadence is 1,178.234792 km. The spectrophotometer Q3 position is 1,174.608319 km. Their difference is 3.626473 km, or 0.308739 percent relative to Q3.

Unlike $H_8/d14C$, this is not a relation between two fixed fitted spacings. The Kepler cadence distance varies with trajectory velocity and passes through the exact Q3 value at one epoch. The relation is therefore a sampling-phase crossing.

14.3 Complete scale hierarchy

Scale	Value km	Role
d14C	1,566.144426	Spectrophotometer primitive equal-phase spacing
3d14C/4	1,174.608319	Spectrophotometer Q3 landmark
H8/2	114,325.000000	Kepler 8-fold half-period
H8	228,650.000000	Kepler 8-fold period
Kepler reciprocal refined	636,407.837954	(1,-2,-3) interplanar spacing
Kepler reciprocal exact	673,775.299475	(1,-2,-3) interplanar spacing
Hsheet	655,343.000000	Rank-2 sheet stacking interval
L2 sheet	2,244,300.000000	Rank-2 sheet in-plane basis
L1 sheet	2,258,266.000000	Rank-2 sheet in-plane basis
Kepler L refined	3,739,968.360000	Direct-lattice scale
Kepler L exact	3,781,125.000000	Direct-lattice scale

Table 19. Spatial scales retained in the combined graph. Values are not collapsed into one universal spacing.

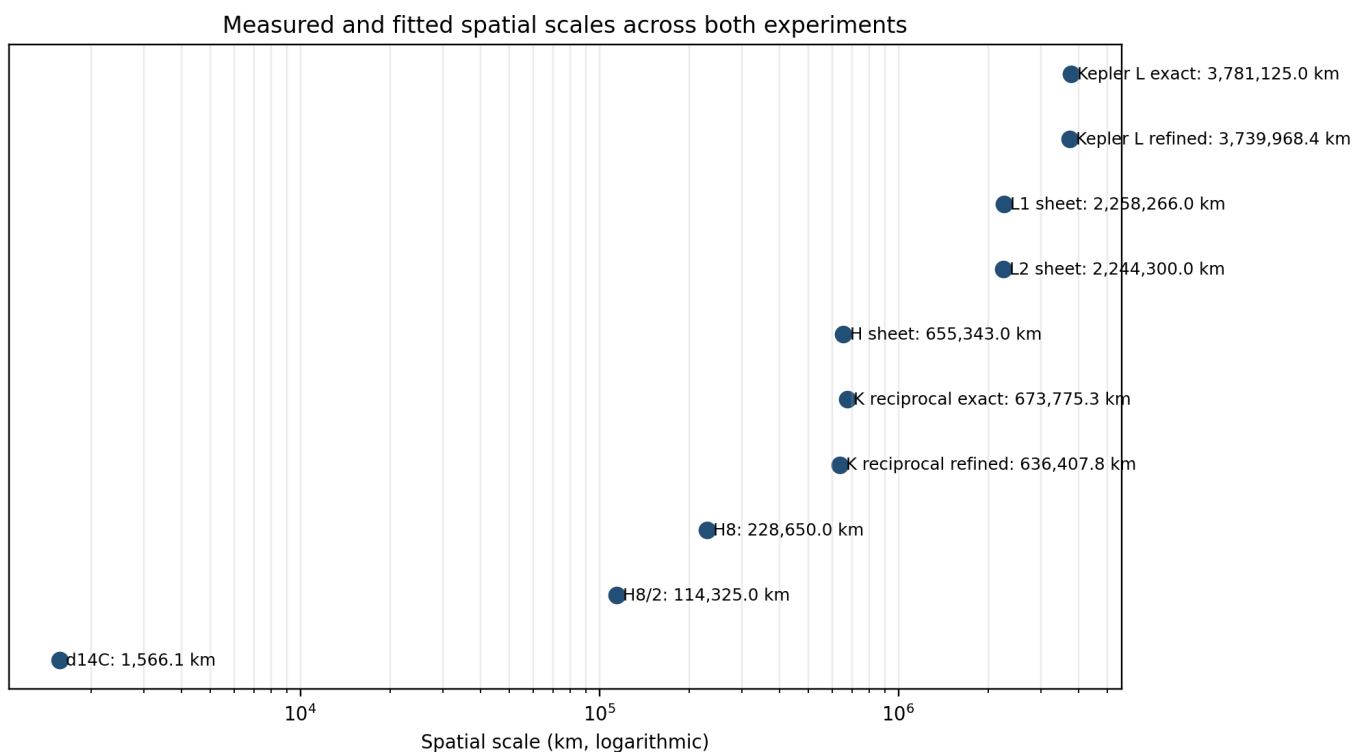


Figure 17. Logarithmic scale ledger across the experiments. The plot shows the hierarchy from the 1,566 km spectrophotometer spacing through the multi-million-kilometre Kepler direct-lattice scale.

15. Side-by-Side Experimental Comparison

Property	Spectrophotometer	Kepler
Platform	Earth laboratory, rotating AS7341 spectrophotometer	Kepler spacecraft pixel data
Primary state	Ten wavelength-channel intensities	Pixel-pair differential variance
Internal state coordinate	Spectral principal components, channel phase, angular harmonics	Complex X/Y variance phase and angular velocity
Trajectory frame	CMB translation + Earth rotation + laboratory + device rotation	Sun-centered J2000 ecliptic spacecraft ephemeris
Resolved direction	q14C; qA-qB transverse frame; qC and qF branches	Translated-lattice reciprocal families; 8-fold grid axes; sheet basis
Reversal observable	Blue versus lower-energy/NIR spectral polarity	CW/CCW angular-rate chirality
Velocity effect	Time period contracts as $q \cdot v$ increases	Distance per cadence contracts as projected speed decreases
Direct geometry	319.393 km x 1,039.778 km exposed base face	Multi-million-km skew direct cell and rank-2 sheet array
Common relation	q14C mixed inside qA-qB plane	(1,-2,-3) reciprocal normal close to q14C
Unresolved coordinate	Longitudinal direct-space period	Common absolute phase with terrestrial experiment

Table 20. Robust side-by-side comparison of the two experimental programs.

16. Unified Spatial-Phase Representation

16.1 Minimum sufficient geometry

The minimum structure capable of containing all observations is a set of reciprocal vectors $\{q_{\mu}\}$. Each defines a spatial phase

$$\Phi_{\mu}(R) = 2\pi q_{\mu} \cdot R + \phi_{\mu}.$$

An observer converts this spatial structure into a temporal phase rate through

$$d\Phi_{\mu}/dt = 2\pi q_{\mu} \cdot v(t).$$

The measured output depends on detector orientation and internal state:

$$Y_D(t) = F_D(\Phi_1(t), \Phi_2(t), \dots; \Omega_D, S_D).$$

Ω_D represents mechanical orientation and rotation. S_D represents the detector-specific spectral, pixel, variance, or memory state. F_D is the transduction rule. The geometric phase can therefore be common while the measured waveform, polarity, and reversal landmarks differ by detector.

16.2 Spectrophotometer transduction

For the spectrophotometer, the measured state is multiplex. Broadband common intensity, blue-versus-lower-energy/NIR polarization, channel-specific phase, quadrature loops, device-angle harmonics, and stable channel pairs all carry different projections of the reciprocal phase coordinates. The 24-hour record demonstrates that one observable can recover qA while another independently recovers qB.

Device-angle pair	Mean coherence m1-m3	m1 phase difference deg	m3 phase difference deg
415-NIR	0.236725	0.544	5.066
445-clear	0.617792	0.884	-5.981
480-630	0.560886	2.291	-2.911
515-680	0.587068	0.700	-7.251
555-590	0.604671	-1.131	-2.652

Table 21. Stable channel-pair graph in pic_id14C. This device-angle layer coexists with the slower spatial phase coordinate.

16.3 Kepler transduction

For Kepler, the measured state is the rotation of a two-component pixel-variance vector. The sign of its angular velocity defines chirality. Different runs of chirality persist for integer numbers of spacecraft cadences. The spatial distance associated with those runs is then determined by projected velocity. This produces discrete run-length bands in distance space without requiring the one-cadence spatial distance to be constant.

16.4 Explanatory compression

One mechanism class explains the following facts simultaneously: nonstationary temporal frequency, coherent spatial phase, detector-specific polarity, multiple reciprocal directions, a common directional sector across independent platforms, integer cadence state durations, and hierarchical scale recurrence. Separate unrelated time oscillators would require a new clock, phase rule, and orientation rule for each instrument and each trajectory segment. The spatial-phase representation replaces those independent rules with fixed geometry plus observer motion and detector response.

17. Observed Data, Derived Results, Inferred Structure, and Proposed Mechanism

Evidence level	Content
Observed	Spectrophotometer counts, Kepler pixel-variance states, timestamps, mechanical indices, spacecraft positions, chirality signs
Derived	Principal components, complex phase, wavevectors, coordinate rotations, interplanar spacings, base-face nodes, projected velocities, cadence run lengths
Mathematical consequence	$d=1/ q $; $d\Phi/dt=2\pi q \cdot v$; phase distance equals integrated normal velocity when definitions are consistent
Cross-domain recurrence	q14C near Kepler (1,-2,-3); q14C inside qA-qB plane; Kepler reciprocal spacing near Hsheet; H8 near 146 d14C
Inferred structure	Multi-axis, hierarchically scaled reciprocal phase architecture
Proposed mechanism class	Detector-specific function F_D that maps common spatial phases, orientation, and internal state into spectral or pixel-variance observables
Open	Longitudinal terrestrial period, common absolute phase registry, microscopic transduction law, higher-precision H8 reconstruction

Table 22. Explicit separation of evidence and inference.

18. Implications for Experimental Science

The central methodological change is the replacement of closed local-cycle analysis by an unwrapped inertial trajectory. A device may return to the same laboratory angle while occupying a new spatial phase because Earth and the spacecraft continue to translate and rotate. Collapsing those samples onto one local circle removes the coordinate required to expose the observed chirp and phase progression.

A second change is the replacement of independent scalar channels by a multiplex state. Spectral channels and pixel pairs form coherent vectors with phase, handedness, quadrature, and matched-pair relationships. The relevant measured object is therefore not one intensity trace but a state evolving through reciprocal phase coordinates.

A third change is the separation of fixed geometry from trajectory-dependent sampling. Spacing and normal are geometric. Time period, distance per cadence, crossing order, and event density depend on observer velocity. This distinction explains why the same spatial organization can appear as a chirp in one experiment and as changing distance bands in another.

The combined framework provides a direct research pathway: recover reciprocal directions from independent state observables; express all directions in one inertial frame; reconstruct direct-space duals where possible; test no-refit phase predictions along new trajectories; and determine the detector-specific transduction law.

19. Falsifiable Predictions and Research Pathway

Test	Operation	Direct prediction
Second-half pic_id14E reconstruction	Use only data outside the pic_id14C interval	qA-qB frame persists with comparable directions and base-face dimensions
Reverse device rotation	Repeat with identical epoch coverage and opposite motor direction	Spatial phase remains tied to inertial trajectory while device-angle carrier reverses
Changed laboratory latitude or axis	Alter Earth-rotation projection on fixed q directions	Temporal chirp changes according to q dot v without changing recovered inertial direction
Kepler no-refit projection	Project original Kepler ephemeris onto fixed q14C and fixed Kepler (1,-2,-3) normals	Chirality events organize in a common phase ledger without changing geometry
Third reciprocal coordinate	Add a trajectory with materially different CMB-parallel velocity	Longitudinal period and closed 3D direct cell become recoverable
Independent detector technology	Measure two or more internal state channels along a known trajectory	Different F_D but recurring reciprocal directions and velocity-controlled phase rate
Higher-precision H8 fit	Refit the chirality branch with more stored digits and fixed event ledger	Determine whether H8/d14C remains near 146 and quantify the true residual

Table 23. Falsifiable next measurements derived from the combined structure.

20. Limitations and Unresolved Dimensions

Boundary	Current status
Record dependence	pic_id14C is nested within pic_id14E; their relation is internal refinement, not independent replication
Transverse-only direct closure	pic_id14E resolves a base face but not the CMB-parallel direct period
Absolute phase origin	Terrestrial and Kepler positions are not yet indexed to one common Sun-centered phase origin
Kepler branch multiplicity	Direct-lattice, sheet-array, 8-fold z, and 98-deg projection branches are distinct and must not be merged
Processing aperture	Kepler chirality uses a 31-sample moving standard deviation; the 31-34 cadence family may reflect state-estimation dynamics
Rounded parameter	H8 is stored as 228,650 km and cannot support a more precise integer-ratio residual than the stored precision allows
Microscopic mechanism	The geometry and kinematics do not yet specify how spatial phase modulates semiconductor emission or pixel variance
External replication	An independent terrestrial acquisition and an independent non-Kepler platform remain required to map generality

Table 24. Explicit limits of the current research graph.

21. Conclusion

The spectrophotometer and Kepler experiments resolve related spatial organization through different detector states. The 24-hour spectrophotometer record exposes a CMB-transverse reciprocal basis qA-qB. The earlier one-axis q14C solution lies within that basis and is reconstructed as a mixed reciprocal coordinate. An independently fitted Kepler direct lattice contains a low-order reciprocal family whose normal is within 0.640396 deg of q14C in the common heliocentric J2000 ecliptic frame. Kepler additionally resolves a rank-2 sheet scale, an 8-fold chirality coordinate, and an integer-cadence reversal grammar. These additional branches show that the common direction is part of a larger multi-axis hierarchy rather than one globally sufficient plane family. Across both experiments, spatial phase rate is controlled by observer velocity, while detector-specific internal states determine the measured spectral or chirality response. The resulting foundational model is a multi-axis, hierarchically scaled spatial-phase architecture sampled by moving orientation-sensitive detectors. The immediate research priority is a no-refit common phase ledger using the original Kepler ephemeris and the second, non-overlapping half of the pic_id14E spectrophotometer record.

Foundation thesis. The extended spectrophotometer record resolves a CMB-transverse reciprocal basis; the shorter record resolves a mixed wavevector within that basis; a Kepler reciprocal-plane family independently recovers nearly the same mixed normal; additional Kepler axes reveal chirality and larger hierarchical periods; and both experiments obey the same trajectory-to-phase-rate grammar.

22. Data Support Section

Item	Value	Status	Source
pic_id14C measurements	84330	Observed	S1
pic_id14C complete turns	937	Observed	S1
pic_id14E measurements	169555	Observed	S4
q14C spacing km	1566.144425556	Fitted/derived	S1
q14C ecliptic lon/lat	303.305802120, +56.925666655	Coordinate conversion	S1
qA spacing km	317.864281381	Fitted	S4
qB spacing km	1034.800486731	Fitted	S4
qA x qB to CMB deg	0.036385969	Derived	S4
q14C from qA-qB residual	0.001480070074	Derived	S1+S4
Kepler refined normal angle to q14C deg	0.640396021	Derived	S6+S1
Kepler refined reciprocal spacing km	636407.837954246	Derived	S6
Kepler sheet H km	655343.000000	Fitted	S7
Kepler H8 km	228650.000000	Fitted	S9
Kepler reversal events	2292	Observed	S8
Kepler base cadence s	58.845500945790	Derived	S8
H8/d14C	145.995475429310	Derived	S1+S9
Terminal kinematic closure m	1.681950591505	Derived	S2-S3

Table 25. Compact numerical support ledger.

23. Reproducibility Notes

- 1 Load S1 and extract q14C, the ICRS normal, plane spacing, channel phases, frequency-track relation, and reversal summary.
- 2 Rotate q14C and all spectrophotometer directions into J2000 ecliptic coordinates using $R_x(-23.439291111 \text{ deg})$.
- 3 Load S2 and independently verify $x_phase = d14C \text{ Delta Phi}$ and x_v from cumulative trapezoidal integration of v_n .
- 4 Load S4 and extract qA, qB, qC, qF, direct vectors a and b, the 16-node display patch, and the plane ledger.
- 5 Solve $[qA \ qB][\alpha \ \beta]^T = q14C$ by least squares and compute residual magnitude and angle from the qA-qB plane.
- 6 Construct exact and refined Kepler direct bases from the declared L, yaw, tilt, skew1, and skew2 values.
- 7 Compute $g_h = A^{-T}(1, -2, -3)$, normalize it, and calculate $d_h = 1/|g_h|$ for both Kepler solutions.
- 8 Compare Kepler and spectrophotometer axes with the axial angle $\text{acos}(|n1 \text{ dot } n2|)$.
- 9 Load S8; compute Delta t, choose the minimum positive cadence, determine $N = \text{round}(\text{Delta } t / \text{Delta } t0)$, and verify microsecond timing closure.
- 10 Compute one-cadence distance $|\text{Delta } z|/N$ and projected velocity $|\text{Delta } z|/\text{Delta } t$. Compare its trajectory sweep with $3d14C/4$.
- 11 Construct the sheet basis from S7 and enumerate nodes $r_s + B[i,j,k]^T$ for i,j,k in $\{-1,0,1\}$.
- 12 Generate figures with equal physical axis scales and label every base-face, direct-cell, and sheet-array node.

23.1 Coordinate and angle equations

$$\text{axial_angle}(n1,n2) = \text{acos}(|n1 \text{ dot } n2|).$$

$$\text{angle_to_plane}(v,n) = \text{asin}(|v_hat \text{ dot } n_hat|).$$

$$\text{distance_from_phase} = d \text{ Delta Phi_cycles}.$$

$$\text{distance_from_velocity} = \sum 0.5(v_i + v_{(i+1)}) \text{ Delta } t_i.$$

$$\text{direct dual: } Q^T D = I, \text{ where } Q=[qA \ qB] \text{ and } D=[a \ b].$$

Appendix A. Complete pic_id14E Base-Face Node Patch

Node	i	j	x km	y km	z km	qA cycles	qB cycles
N-1_-1	-1	-1	-344.469340	494.637081	-868.856661	-1.000000000	-1.000000000
N-1_+0	-1	0	-23.580892	-241.209359	-208.024245	-1.000000000	-0.000000000
N-1_+1	-1	1	297.307557	-977.055799	452.808171	-1.000000000	1.000000000
N-1_+2	-1	2	618.196005	-1712.902239	1113.640586	-1.000000000	2.000000000
N+0_-1	0	-1	-320.888448	735.846440	-660.832416	0.000000000	-1.000000000
N+0_+0	0	0	0.000000	0.000000	0.000000	0.000000000	0.000000000
N+0_+1	0	1	320.888448	-735.846440	660.832416	-0.000000000	1.000000000
N+0_+2	0	2	641.776897	-1471.692880	1321.664831	-0.000000000	2.000000000
N+1_-1	1	-1	-297.307557	977.055799	-452.808171	1.000000000	-1.000000000
N+1_+0	1	0	23.580892	241.209359	208.024245	1.000000000	0.000000000
N+1_+1	1	1	344.469340	-494.637081	868.856661	1.000000000	1.000000000
N+1_+2	1	2	665.357789	-1230.483521	1529.689076	1.000000000	2.000000000
N+2_-1	2	-1	-273.726665	1218.265158	-244.783926	2.000000000	-1.000000000
N+2_+0	2	0	47.161783	482.418718	416.048490	2.000000000	0.000000000
N+2_+1	2	1	368.050232	-253.427722	1076.880906	2.000000000	1.000000000
N+2_+2	2	2	688.938680	-989.274162	1737.713321	2.000000000	2.000000000

Table A1. Complete dimensionally defined 4 x 4 base-face patch.

Appendix B. pic_id14E Plane Equations

Family	Index	Spacing km	Offset km	Equation in metres
A	-1	317.864281	-317.864281	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = -1$
A	0	317.864281	0.000000	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 0$
A	1	317.864281	317.864281	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 1$
A	2	317.864281	635.728563	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 2$
B	-1	1034.800487	-1034.800487	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = -1$
B	0	1034.800487	0.000000	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 0$
B	1	1034.800487	1034.800487	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 1$
B	2	1034.800487	2069.600973	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 2$

Table B1. Stored A and B plane-family equations.

Appendix C. Kepler Direct-Cell Vertex Coordinates

Exact solution

Node	i	j	k	x ecliptic km	y ecliptic km	z ecliptic km
V000	0	0	0	0.000	0.000	0.000
V001	0	0	1	-3589479.135	1099535.086	-451185.552
V010	0	1	0	-35717.767	3753900.444	451510.755
V011	0	1	1	-3625196.901	4853435.531	325.203
V100	1	0	0	-3016783.767	474411.051	2229541.685
V101	1	0	1	-6606262.901	1573946.137	1778356.133
V110	1	1	0	-3052501.534	4228311.495	2681052.440
V111	1	1	1	-6641980.668	5327846.581	2229866.888

Table C. Exact solution eight-corner direct cell.

Refined solution

Node	i	j	k	x ecliptic km	y ecliptic km	z ecliptic km
V000	0	0	0	0.000	0.000	0.000
V001	0	0	1	-3475961.638	1339702.856	-332039.578
V010	0	1	0	44931.162	3704426.693	512413.313
V011	0	1	1	-3431030.476	5044129.549	180373.734
V100	1	0	0	-2991666.493	637749.382	2151876.079
V101	1	0	1	-6467628.131	1977452.238	1819836.500
V110	1	1	0	-2946735.331	4342176.075	2664289.391
V111	1	1	1	-6422696.969	5681878.931	2332249.813

Table C. Refined solution eight-corner direct cell.

Appendix D. Complete Kepler Sheet-Array 27-Node Neighborhood

Node	i	j	k	x km	y km	z km
S-1-1-1	-1	-1	-1	-49559.162	-1170724.964	-1405146.638
S-1-1+0	-1	-1	0	-392554.590	-687144.757	-1684393.423
S-1-1+1	-1	-1	1	-735550.019	-203564.550	-1963640.208
S-1+0-1	-1	0	-1	640062.855	242886.257	195792.861
S-1+0+0	-1	0	0	297067.426	726466.465	-83453.924
S-1+0+1	-1	0	1	-45928.002	1210046.672	-362700.709
S-1+1-1	-1	1	-1	1329684.871	1656497.479	1796732.360
S-1+1+0	-1	1	0	986689.442	2140077.686	1517485.575
S-1+1+1	-1	1	1	643694.014	2623657.893	1238238.790
S+0-1-1	0	-1	-1	-93369.218	-65031.269	563428.344
S+0-1+0	0	-1	0	-436364.646	418548.938	284181.559
S+0-1+1	0	-1	1	-779360.075	902129.145	4934.774
S+0+0-1	0	0	-1	596252.798	1348579.952	2164367.843
S+0+0+0	0	0	0	253257.370	1832160.159	1885121.058
S+0+0+1	0	0	1	-89738.059	2315740.366	1605874.273
S+0+1-1	0	1	-1	1285874.814	2762191.173	3765307.342
S+0+1+0	0	1	0	942879.386	3245771.380	3486060.557
S+0+1+1	0	1	1	599883.958	3729351.588	3206813.772
S+1-1-1	1	-1	-1	-137179.274	1040662.425	2532003.326
S+1-1+0	1	-1	0	-480174.703	1524242.633	2252756.541
S+1-1+1	1	-1	1	-823170.131	2007822.840	1973509.756
S+1+0-1	1	0	-1	552442.742	2454273.647	4132942.825
S+1+0+0	1	0	0	209447.313	2937853.854	3853696.040
S+1+0+1	1	0	1	-133548.115	3421434.061	3574449.255
S+1+1-1	1	1	-1	1242064.758	3867884.868	5733882.324
S+1+1+0	1	1	0	899069.330	4351465.075	5454635.539
S+1+1+1	1	1	1	556073.901	4835045.282	5175388.754

Table D1. All 27 nodes displayed in Figure 14.

Appendix E. pic_id14C Channel and Pair Ledgers

Channel	Plane-wave R2	Amplitude	Phase deg	Null percentile
415	0.372062	0.740841	242.975	100.000
445	0.457417	0.652038	238.230	100.000
480	0.601312	1.251609	253.054	100.000
515	0.123141	0.173917	30.955	97.656
555	0.230059	0.387285	71.003	100.000
590	0.138933	0.193531	91.164	98.438
630	0.206067	0.335646	56.471	98.438
680	0.107285	0.163760	53.226	88.281
NIR	0.540354	0.904296	82.014	100.000
clear	0.059378	0.197769	298.433	67.969

Table E1. Complete ten-channel plane-wave fit ledger.

Matched pair	Mean coherence	m1 phase deg	m3 phase deg
415-NIR	0.236725	0.544	5.066
445-clear	0.617792	0.884	-5.981
480-630	0.560886	2.291	-2.911
515-680	0.587068	0.700	-7.251
555-590	0.604671	-1.131	-2.652

Table E2. Complete stable harmonic-pair ledger.

Appendix F. Selected Kepler Reversal Rows

Flip	BJD	z grid km	z mod H km	z mod H/2 km	New sign
1	260.252816511533	26881291.409894	84888.137434	84888.137434	1
2	260.280740881197	26828085.563804	31682.291343	31682.291343	-1
3	260.282784127758	26824192.617869	27789.345408	27789.345408	1
4	260.283465209944	26822894.974222	26491.701762	26491.701762	-1
5	260.284827374318	26820299.694427	23896.421966	23896.421966	1
6	260.285508456505	26819002.058279	22598.785818	22598.785818	-1
7	260.286189538692	26817704.424630	21301.152170	21301.152170	1
8	260.328416634282	26737256.028236	169502.755775	55177.755775	-1
2285	289.894875451075	-24762306.718626	116190.008914	1865.008914	1
2286	289.896918697636	-24765366.037435	113130.690104	113130.690104	-1
2287	289.900324108570	-24770464.699582	108032.027957	108032.027957	1
2288	289.928929560421	-24813283.464588	65213.262951	65213.262951	-1
2289	289.939145793225	-24828571.548942	49925.178597	49925.178597	1
2290	289.941870121973	-24832647.986270	45848.741269	45848.741269	-1
2291	289.957535012272	-24856084.352990	22412.374549	22412.374549	1
2292	289.960259341020	-24860159.695302	18337.032237	18337.032237	-1

Table F1. First and final eight rows of the 2,292-event ledger.

Appendix G. Source Integrity Manifest

File	Bytes	SHA-256
K-Field_spectrophotometer_pic_id14C_analysis_1785888480.js on	2929454	5f5c091ba05fab9a5ea5ed0bc78d2673dac82444452f7eb6ee5dbedbbe37df0
K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane _structure_1785932820.json	1545939	79e3aa2b187a9ad4fbd2b481de1b22ff7c53753c738678f3f593d3c0fc8e3c85
pic_id14C_plane_normal_kinematics_timeseries.csv	199811	26319076c3bc2f9c8106f513983dec13dad6de27bcad206b70e3390bdb90a0a8
pic_id14C_plane_normal_kinematics_summary.json	1193	ea16e27301bee285ea2117d6eebb5946bfc8e22728624e50cfc5f5c1ca5b609d
polarity_flips_vs_z_axis.csv	190776	d52072e7fc9f25716b624a993a46be5a45d7ace85f8942f62098cf8d3398436b
results_new_translation2_stride_2.txt	2605792	e50caf52faca814735ef0f449b6dcfd486def1895cf94032aac856c964804831
kepler_lattice_plane_fit_report_v1.pdf	263843	2ba6e11f35000f4d56081840f0048ae57e159576a9f37db8d1d57b2937114c8f
sheet_lattice_final_paper.pdf	394679	6018a056999116ce54fbbc930903976ecaf3a62ce8aef9d6abaf5116485a4f7d
fit_params.json	109	2375dc1ed33acec870f7bc6d8b70bdb0603a58a57037a6b0602b30ef42f7f406
README(1).txt	1423	ca41e82e747be6fe3a10ef93d48dcdb63da2ead14bca3cb3c7bc213754b3e430
forbidden_z_steps_matrix_100km(1).csv	11734	361bc3ccacd2a612cae0ac6775a26633000c0a39c759921bd958de0bbd0ad518
K-Field_spectrophotometer_resolved_earth_traversal_through_ rectilinear_plane_system_1785930261.pdf	1084936	c554ed8602a888898dd248e6a008f184f9e96e8c5502afab1992764d2a26ec7d

Table G1. Source files used to construct this report. Hashes identify the exact byte streams available in the working archive.

Appendix H. Machine-Readable Companion Schema

The companion file `K-Field_spectrophotometer_Kepler_comparison_1785936602.json` contains report metadata, source hashes, the full derived comparison ledger, exact coordinate vectors, direct and reciprocal bases, complete base-face nodes, Kepler cell vertices, sheet-array nodes, event statistics, figure manifest, and equation registry. Numeric arrays are stored in SI units or explicitly labeled kilometre units.

The companion is designed for audit, plotting, independent recalculation, and future no-refit cross-projection studies.